

Global Climate and Wildfire Data Visualization

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This report documents our STATS 401 final project, *Global Climate and Wildfire Data Visualization*. Our goal is to investigate how wildfire detections co-evolve with major climate indicators through a coordinated five-view interactive dashboard. We combine global wildfire detections by source type, sampled geospatial wildfire points, global CO2 totals, global temperature and precipitation summaries, country-level CO2 and CO2-attributed temperature change, and wildfire-related social text terms with sentiment scores. The implementation is built with D3.js and organized into five linked pages with rich interaction: metric switching, timeline playback, speed control, geographic filtering, dual-handle year-range brushing, and dynamic text exploration. At least one complex representation is included via an interactive rotating globe and geospatial map pair. The final system supports temporal trend analysis, spatial hotspot inspection, country-level comparison, climate–wildfire correlation reading, and qualitative interpretation of wildfire discourse. We discuss strengths, limitations, and concrete improvements for future work.

Additional Key Words and Phrases: wildfire, climate indicators, D3.js, interactive visualization, correlation heatmap, rotating globe, word cloud

1 Introduction

Wildfire activity has become a global concern because of ecosystem damage, human loss, and climate feedback effects. At the same time, climate indicators such as CO2 emissions, annual temperature, and precipitation continue to change across years. Our project asks whether and how these large-scale climate signals move with observed wildfire detections, and how the relationships differ across temporal, spatial, country-level, and text perspectives.

Our visualization goals are:

- (1) show long-term wildfire detection trends and compare them with climate trends;
- (2) provide global spatial context through both a flat map and a rotating globe;
- (3) support country-level comparison between CO2 emissions and CO2-attributed temperature change;
- (4) quantify pairwise climate–wildfire associations over adjustable time windows;
- (5) summarize wildfire-related language and sentiment patterns in social text.

The full dashboard is implemented in D3.js and published as five coordinated interactive views. Figure 1 shows the dashboard introduction page and project scope.

2 Method

2.1 Data sources

We used four primary data families and one derived text dataset:

- **Wildfire detections (global, 2012–2025):** NASA FIRMS archives (VIIRS fire archive CSVs), documented in `data/wild_fire.ipynb`. Download entry: [NASA FIRMS download portal](#).
- **Wildfire social-media dataset (CrisisMMD):** CrisisMMD multimodal crisis dataset (wildfire-related subsets used for text analysis). Dataset page: [CrisisMMD dataset page](#).
- **Climate summaries (temperature and precipitation):** NASA POWER monthly/annual export used in `data/weather.ipynb`, API example: [NASA POWER API example](#).
- **CO2 and country climate indicators:** OWID CO2 dataset used in `data/co2.ipynb` and Visualization 3. Dataset page: [OWID CO2 dataset page](#); CSV endpoint: [OWID CO2 CSV endpoint](#).

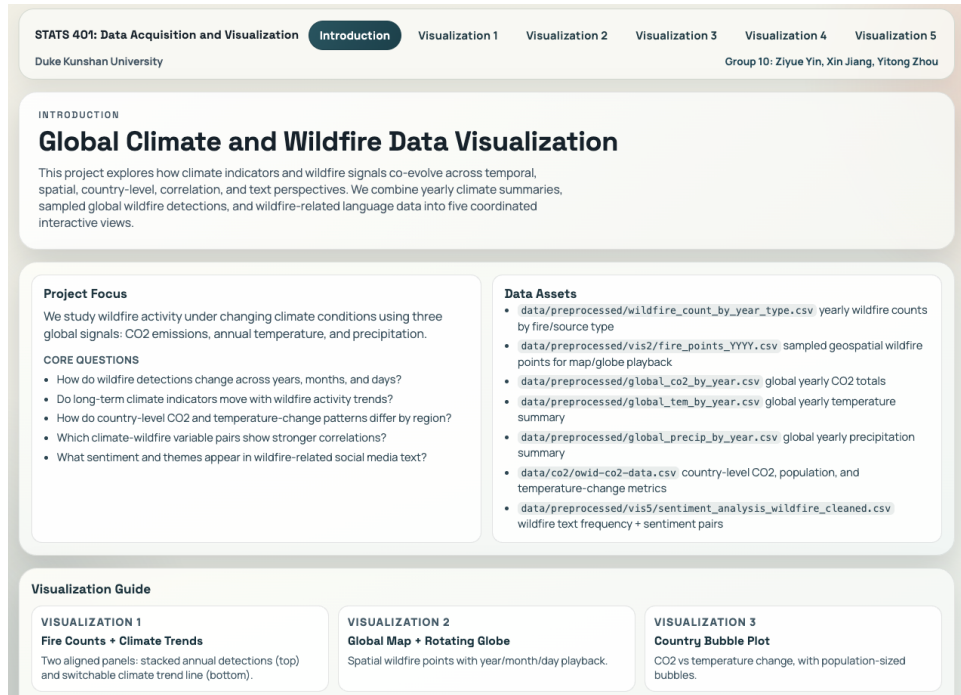


Fig. 1. Introduction page of the final dashboard, including project focus, data assets, and visualization guide.

For Visualization 5, the repository contains only the cleaned term-level table (no raw corpus notebook). Based on corpus vocabulary and wildfire context, we additionally provide a public wildfire social-media data reference used in related work: Tubbs Fire Twitter dataset (IEEE DataPort), linked from <https://www.frontiersin.org/journals/big-data/articles/10.3389/fdata.2024.1369878/full> and hosted at <https://ieee-dataport.org/open-access/tubbs-fire-twitter-dataset>. This is used here as an external provenance link when local notebook provenance is absent.

2.2 Collection, cleaning, and preprocessing pipeline

Wildfire aggregation. In `data/wild_fire.ipynb`, yearly FIRMS archive CSVs are read in chunks to control memory usage. We keep `acq_date` and `type`, parse year from date, and aggregate counts by `(year, type)` into `data/preprocessed/wildfire_count_by_year_type.csv`.

Global CO2 aggregation. In `data/co2.ipynb`, OWID rows are filtered to ISO3 country entries, numeric fields are cleaned, and global annual CO2 totals are computed by summation over countries. We also compute a population-weighted global per-capita CO2 series, then export `data/preprocessed/global_co2_by_year.csv`.

Climate yearly summaries. Temperature and precipitation annual values are read from NASA POWER export files and stored as `global_tem_by_year.csv` and `global_precip_by_year.csv`. For the dashboard overlap with wildfire years, the effective climate range is 2012–2024.

Vis2 sampling for interactive rendering. The script `scripts/build_vis2_fire_samples.py` performs reservoir sampling (`SAMPLE_SIZE=15000` per year) from raw yearly FIRMS files. It validates latitude/longitude ranges, keeps

key fields, outputs `fire_points_{year}.csv`, and writes `sample_summary.csv`. This step makes map/globe playback responsive in browser-side D3.

Vis3 filtering and mapping. Visualization 3 loads OWID country-year data and `country_to_region.csv`, then filters to valid numeric `co2`, `temperature_change_from_co2`, and `population>0`. Region is encoded by color and defaults to “Other” when a mapping is missing.

Vis4 correlation computation. Visualization 4 builds yearly aligned vectors for 7 variables (CO2, temperature, precipitation, vegetation fire, volcano, static land, offshore) and computes Pearson correlation:

$$r = \frac{n \sum xy - \sum x \sum y}{\sqrt{(n \sum x^2 - (\sum x)^2)(n \sum y^2 - (\sum y)^2)}}.$$

The correlation matrix is recomputed whenever the selected year range changes.

Vis5 text encoding. Visualization 5 loads word-level frequencies and sentiment values. Sentiment thresholds are defined in code as positive > 0.05 , neutral $[-0.05, 0.05]$, negative < -0.05 . Word size uses square-root scaling of frequency, and color encodes sentiment category.

2.3 Data report

We summarize the core datasets in text form below. The wildfire detection summary file `wildfire_count_by_year` contains 56 rows covering 2012–2025 across four fire types, representing 280 396 425.000 raw detections; its attributes include fire type (categorical), year (ordinal), and count (numerical). The sampled point set `fire_points` includes 210 000.000 points (14 years $\times$ 15000), with type (categorical), date (ordinal), and latitude/longitude, FRP, and brightness (numerical). The global CO2 series `global_co2_by_year` spans 275 years (1750–2024) with year (ordinal) and global total CO2 plus weighted per-capita CO2 (numerical). The climate summaries `global_tem_by_year` and `global_precip_by_year` each cover 44 years (1981–2024), with year (ordinal) and annual mean temperature or precipitation (numerical). The country-year table `owid-co2-data` includes 50,411 raw rows and 24,233 rows after Vis3 filtering; its attributes include country/region (categorical), year (ordinal), and CO2, temperature change, and population (numerical). Finally, the term list `sentiment_analysis_wildfire` contains 3,690 terms with word/sentiment class (categorical) and frequency and sentiment score (numerical).

This satisfies the assignment’s scale and attribute-type expectations: the raw wildfire source alone is far above 10,000 items, and the project contains categorical, ordinal, and numerical attributes across views.

3 Results: Five Interactive Visualizations

3.1 Visualization 1: Fire counts + climate trends

Visualization 1 combines stacked yearly wildfire detections (top panel) and a switchable climate line chart (bottom panel) in one aligned time axis. The bars encode four source types by color. The line can switch among global CO2, temperature, and precipitation.

Interactions implemented:

- metric toggle buttons (CO2 / Temperature / Precipitation);
- year-hover shared highlight band and focus line across both panels;
- per-year tooltip showing total detections, type breakdown, and selected climate value.

Figures 2 and 3 show this view under different metric selections. In 2012–2025, total detections range from 17 251 846.000 (2022) to 21 842 028.000 (2023). Climate values are shown through 2024, and the chart explicitly annotates 2025 climate unavailability.

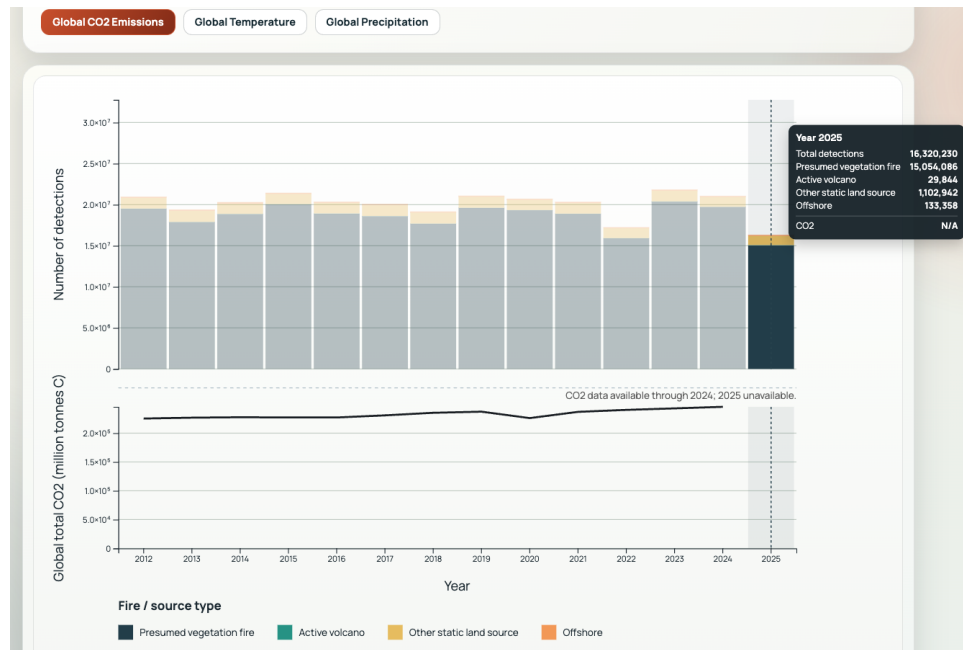


Fig. 2. Visualization 1 with CO2 selected: stacked wildfire detections plus global CO2 trend.

3.2 Visualization 2: Global map + rotating globe (complex view)

Visualization 2 is our complex representation: a dual geospatial view with a flat world map and an orthographic rotating globe rendering the same wildfire points. Data are year-based samples with optional month/day filtering.

Interactions implemented:

- playback scale switch (year, month, day);
- timeline play/pause with speed control;
- year/month/day selectors;
- globe longitude/latitude sliders;
- pause/resume and reset globe rotation;
- direct drag interaction on globe.

The status line reports sampled versus original yearly detection counts, and a type-wise statistics panel updates with current filters. Figure 4 shows the two synchronized spatial views.

3.3 Visualization 3: Country bubble plot

Visualization 3 maps country-level co2 to the x-axis, temperature_change_from_co2 to the y-axis, population to bubble size, and region to bubble color. A large year watermark helps temporal orientation during animation.

Interactions implemented:

- year slider and autoplay;
- playback speed control;
- region multi-select checkboxes;
- country search + country multi-select list;

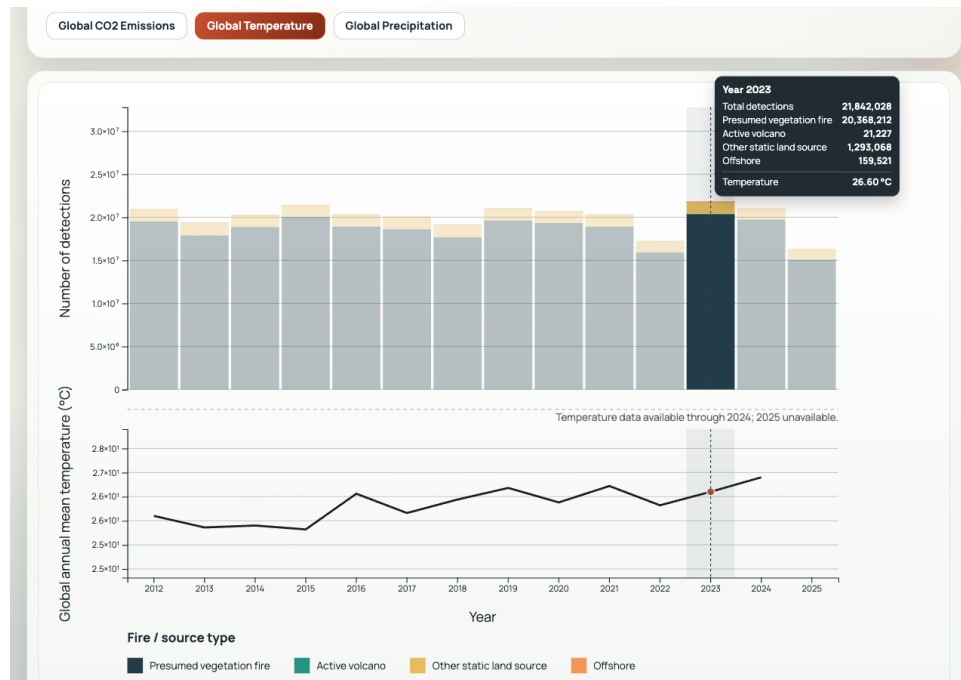


Fig. 3. Visualization 1 with temperature selected (same interaction model).

- select-all / clear-all filters.

For example, in year 1990 the chart shows about 220 countries after filtering; in 2024 it shows 226. Figure 5 illustrates the interactive layout and filter panel.

3.4 Visualization 4: Climate–wildfire correlation heatmap

Visualization 4 computes a 7×7 Pearson correlation matrix over user-selected year ranges. The matrix is split into climate factors and wildfire types, with diverging color encoding from negative (blue) to positive (red), and a “Key Insights” panel listing strongest positive/negative pairs.

Interactions implemented:

- dual-handle year-range slider;
- range playback animation with speed control;
- hover tooltip for each matrix cell;
- auto-refresh of ranked insight cards as range changes.

For the 2012–2024 range, notable results include strong positive CO₂–precipitation (≈ 0.82) and CO₂–temperature (≈ 0.73), and moderate negative temperature–volcano (≈ -0.68). Figure 6 shows the rendered matrix and insight panel.

3.5 Visualization 5: Wildfire text sentiment word cloud

Visualization 5 summarizes wildfire-related text as a word cloud. Word size encodes frequency, and color encodes sentiment category. The current dataset includes 3,690 cleaned terms; top words include *fire* (457), *northern* (114), and *californias* (110).

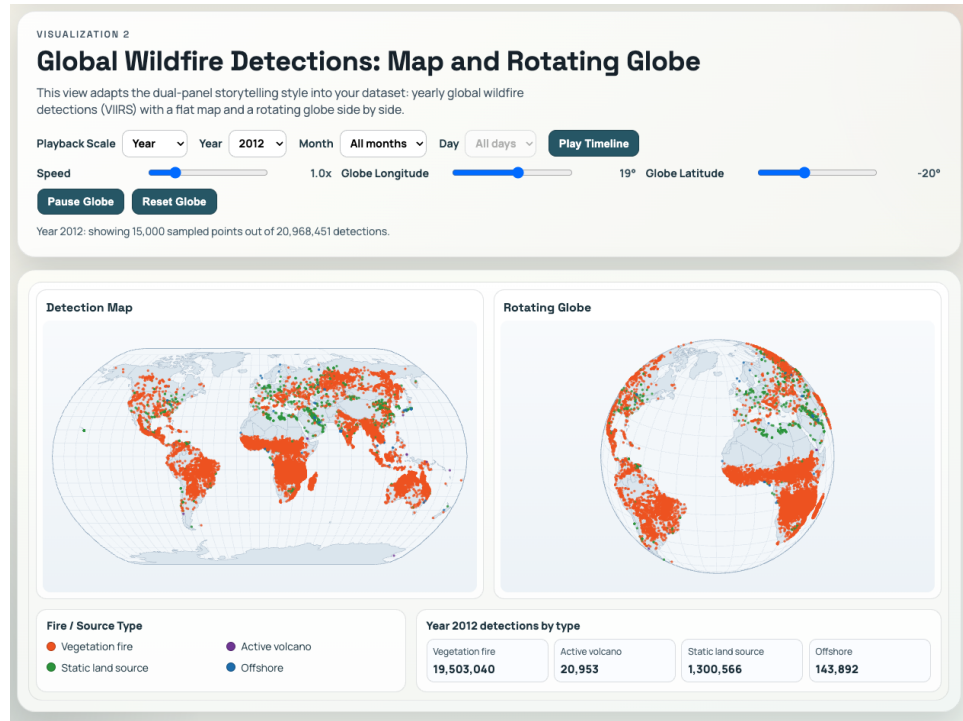


Fig. 4. Visualization 2: synchronized detection map and rotating globe with timeline controls.

Interactions implemented:

- Top- N slider (up to 500 terms);
- hover tooltip with term, frequency, sentiment value and class;
- dynamic summary panel and top-term list synchronized with current N .

Figure 7 and Figure 8 show how term density and context change when Top- N increases.

4 Discussion

4.1 What-Why-How evaluation

What (user tasks). The dashboard supports five complementary tasks: long-term trend comparison (Vis1), spatial hotspot exploration (Vis2), country-level structural comparison (Vis3), multi-variable association analysis (Vis4), and discourse/sentiment interpretation (Vis5).

Why (design rationale). We used distinct visual idioms to match task types:

- stacked bars + line for annual totals and aligned trend reading;
- map + globe for global spatial cognition and rotational perspective;
- bubble chart for multivariate country comparison;
- heatmap for dense pairwise correlation scanning;
- word cloud for theme salience and sentiment cues.

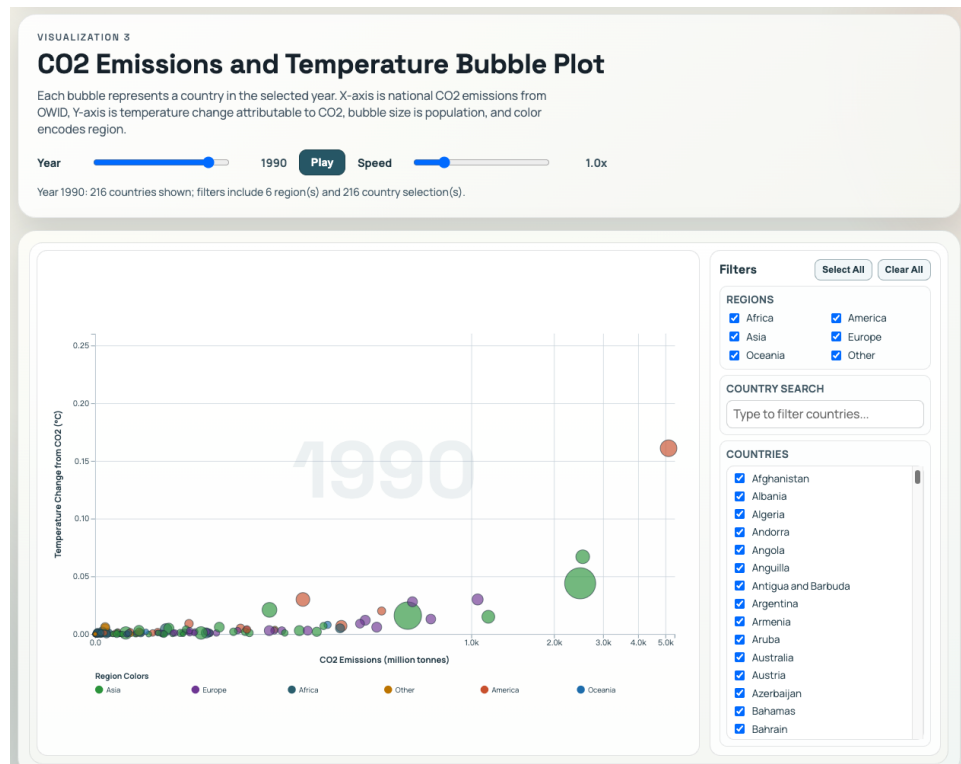


Fig. 5. Visualization 3: CO2–temperature bubble plot with region and country filters.

How (encoding and interaction). All views use explicit channels (position, area, color, text labels) and interaction to reduce cognitive load: direct manipulation sliders, hover tooltips, playback controls, and filter panels. Interaction count is well above the minimum requirement of five.

4.2 Strengths

- **Completeness:** all five required interactive visualizations are implemented and published in one coherent dashboard experience.
- **Technical depth:** Vis2 includes a true orthographic globe with drag, manual rotation, and temporal playback over sampled point clouds.
- **Data integration:** climate, wildfire, country-level indicators, and text sentiment are integrated from multiple sources into coordinated storytelling.
- **Usability:** consistent styling, clear legends, status lines, and context-aware tooltips improve interpretability.

4.3 Limitations and potential resolutions

- **Temporal overlap limits:** Vis4 correlation is effectively computed on 13 years (2012–2024), so coefficients can be sensitive to short-term fluctuation. Future work can use longer aligned climate windows and robust statistics.

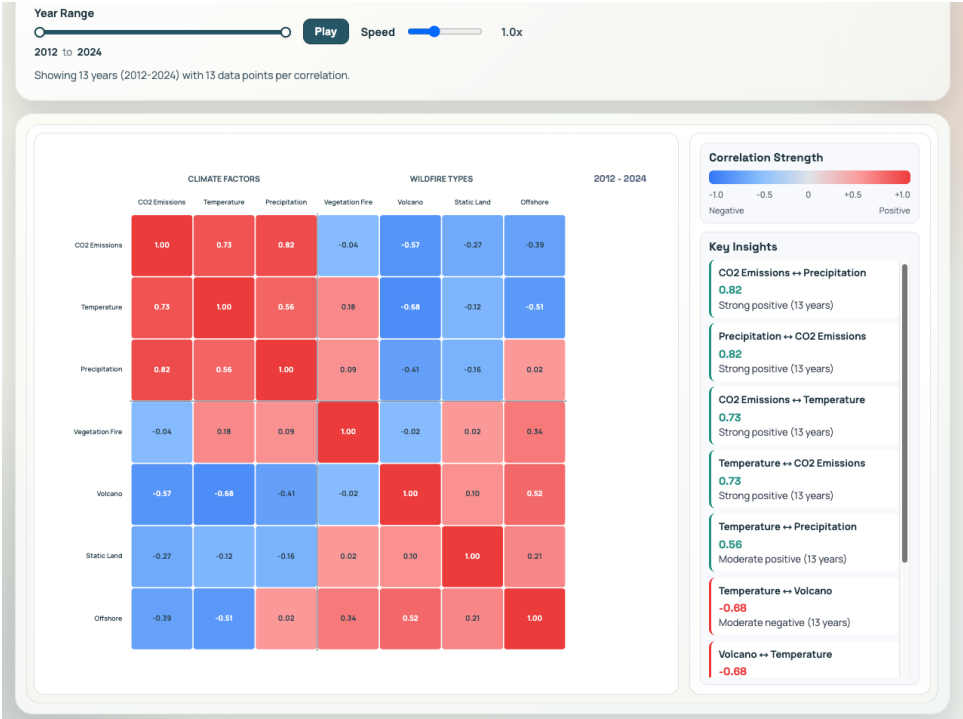


Fig. 6. Visualization 4: interactive climate–wildfire correlation heatmap and ranked insights.

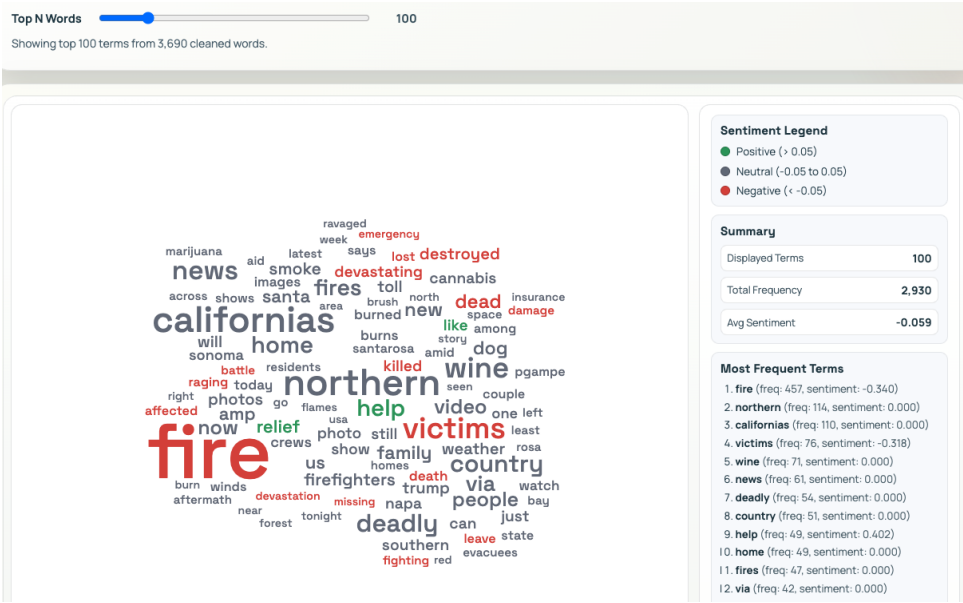


Fig. 7. Visualization 5 with Top 100 terms.



Fig. 8. Visualization 5 with Top 500 terms.

- **Climate representation scope:** current temperature/precipitation series come from a specific NASA POWER export setting and should be expanded to globally gridded, area-weighted products for stronger global interpretation.
- **Sampling trade-off in Vis2:** 15,000 points/year enables smooth interaction but hides fine local density. Multi-resolution sampling and zoom-triggered loading can improve fidelity.
- **Text provenance gap:** the repo currently stores cleaned term-frequency sentiment output rather than full raw text pipeline. A reproducible raw-to-clean NLP notebook should be added in the next iteration.
- **Causality caution:** Vis4 shows correlation, not causation. Additional lag analysis or causal inference methods would strengthen interpretation.

5 Conclusion

This project delivers a full five-view interactive visualization system for exploring climate and wildfire patterns across multiple analytical perspectives. By combining temporal summaries, spatial maps, country-level indicators, correlation structure, and social-text sentiment, we provide a richer interpretation than any single chart can offer.

Our results suggest that CO₂, temperature, and precipitation trends can be explored alongside wildfire detections in a transparent and interactive workflow, while also revealing important limits in data coverage and interpretation. The final dashboard contributes both as an educational visual analytics artifact and as a reusable foundation for future, higher-resolution climate-wildfire analysis.

6 Roles and Responsibilities

- **Ziyue Yin:** project integration, data preprocessing pipeline orchestration, Visualization 1 and 2 implementation, deployment and repository maintenance.

- **Xin Jiang:** Visualization 3 and 4 implementation, interaction logic for filters/range controls, and correlation analysis workflow.
- **Yitong Zhou:** Visualization 5 implementation, interface refinement, narrative writing support, figure preparation, and presentation materials.

Code Availability and Website Access

- Code Link: https://github.com/larry-ziyue-yin/global_climate_and_wildfire_data_visualization
- Website Link: https://larry-ziyue-yin.github.io/global_climate_and_wildfire_data_visualization/